

Reposys Campus Reprography Automation System

Requirement Gathering, User Stories & Proposed UI Design

1. Requirement Gathering & Analysis

The requirement gathering phase for the proposed Reposys Campus Reprography Automation System was conducted by analyzing the physical bottlenecks of traditional campus print shops. The goal is to map these real-world challenges into a highly distributed MERN-stack architecture. The requirements are designed to emphasize database flexibility, real-time communication needs, and mobile-first accessibility.

1.1 Proposed Functional Requirements (FR)

Based on the initial system blueprint, the core functional requirements to be developed are defined as follows:

FR-01 (Authentication & Security): The system shall utilize Nodemailer for OTP-based email verification before allowing users to place orders, ensuring only verified campus emails are registered.

FR-02 (File Processing Pipeline): The system shall offload PDF storage to third-party cloud infrastructure (Cloudinary) to prevent MongoDB bloat, and perform automated page counting prior to rendering print options.

FR-03 (Financial Transactions): The system shall integrate a digital Wallet architecture that supports secure top-ups (via Razorpay) and peer-to-peer "Split Requests" utilizing MongoDB ACID Transactions to prevent data loss.

FR-04 (Real-Time Synchronization): The system shall use WebSockets (Socket.io) to push live order status updates to the client's browser, eliminating the need for manual page refreshes.

FR-05 (Intelligent Queuing): The Staff module shall sort incoming tasks using a Composite Scoring Model rather than standard FIFO, factoring in VIP status, page count, and timestamp.

FR-06 (AI Context Injection): The system shall include a Google Gemini-powered AI Chatbot that is fed live system context (e.g., current A4 prices, queue length) by the backend before generating responses.

1.2 Proposed Non-Functional Requirements (NFR)

NFR-01 (Accessibility): The frontend shall be installable as a Progressive Web App (PWA) with offline capabilities to ensure students can access the system reliably on mobile devices.

NFR-02 (Data Flexibility): The system shall utilize a NoSQL structure to accommodate highly variable print parameters (e.g., varying page ranges, color vs. B&W, binding requirements) without rigid relational table constraints.

2. Comprehensive User Story Documentation

The following 42 user stories have been explicitly defined to guide the upcoming development sprints. They cover every major actor (User, Faculty, Staff, Admin, Guest) and enforce strict acceptance criteria regarding security, state transitions, and asynchronous operations (Socket.io, Nodemailer, Razorpay).

1. Authentication & Account Management

ID	User Story	Acceptance Criteria	Details
AUTH-01	As a Student, I want to register with my college email and ID, so that I can access reprography services.	Email + college ID uniqueness validated; password meets complexity rules; verification email sent; account inactive until verified	P: High C: Medium
AUTH-02	As a Faculty member, I want my registration to require Admin approval, so that role privileges aren't self-assigned.	Faculty signup sets pendingRoleApproval: true; Admin gets notification; user stays Student-level access until approved	P: High C: Medium
AUTH-03	As any registered user, I want to log in securely and stay logged in across sessions, so that I don't re-authenticate constantly.	JWT issued in httpOnly cookie; session persists per device; rate-limited login attempts (10/15min)	P: High C: Medium
AUTH-04	As a user, I want to reset my password if I forget it, so that I'm not locked out.	Reset link emailed, expires in 1 hour, single-use token	P: High C: Low
AUTH-05	As a user, I want to log out of all devices from one place, so that I can secure my account if a device is lost.	"Logout all devices" invalidates all sessions issued before the action timestamp	P: Medium C: Low

2. Document Upload & AI Analysis

ID	User Story	Acceptance Criteria	Details
DOC-01	As a Student, I want to upload PDF/DOC/DOCX/JPG/PNG files, so that I can order print/copy/scan/bind services.	File type + size validated server-side (signature check, not just extension); max 10 documents/order	P: High C: Medium
DOC-02	As a Student, I want the system to auto-detect blank pages and quality issues (too dark/light/low-res), so that I can avoid paying for bad print output.	Each page analyzed; flagged pages shown in preview with badges; user can still proceed regardless	P: High C: High
DOC-03	As a Student, I want to convert DOCX to PDF before uploading, so that formatting is preserved for print.	Conversion runs via backend service; converted file previewed before confirming; fallback to manual upload on failure	P: Medium C: Medium

3. Order Placement

ID	User Story	Acceptance Criteria	Details
ORD-01	As a Student, I want to configure print settings (copies, colour, sides, paper size, binding), so that my order matches my requirements.	Cost recalculates live as options change; server recalculates cost again at order creation (never trusts client value)	P: High C: Medium
ORD-02	As a Faculty member, I want my orders to get priority queue placement over Students, so that my orders process faster given role weighting.	Priority score = role weight + aging factor; Faculty base weight lower (higher priority) than Student	P: High C: Medium
ORD-03	As a Student, I want to see the exact cost breakdown before paying, so that there are no surprises.	Breakdown shows base rate, page count, copies, binding charge, double-sided discount if any	P: High C: Low
ORD-04	As a Student, I want to cancel an order before it starts processing, so that I don't pay for something I no longer need.	Cancel disabled once status = Processing; refund triggered automatically for online payments	P: Medium C: Medium
ORD-05	As a Student, I want to reorder a past order with one click, so that I don't have to reupload and reconfigure everything.	Reorder pre-fills wizard step 3 with same documents and config; user can still edit before confirming	P: Low C: Low

4. Queue & Order Tracking

ID	User Story	Acceptance Criteria	Details
QUEUE-01	As a Student, I want to see my live queue position and estimated wait time, so that I know when to visit the counter.	Wait time = sum of estimatedDuration of all orders ahead in blocking statuses; updates via Socket.io in real time	P: High C: Medium
QUEUE-02	As a Guest, I want to track my order using a public link without logging in, so that I can check status from any device.	Public track route requires no auth; shows only status + token, no sensitive data	P: Medium C: Low
QUEUE-03	As a Student, I want to be notified when my order reaches "Ready for Pickup," so that I know exactly when to collect it.	Push + in-app notification fires on status change; pickup OTP included	P: High C: Low

5. Payment & Wallet

ID	User Story	Acceptance Criteria	Details
PAY-01	As a Student, I want to pay online via Razorpay, so that I don't need cash at the counter.	Payment signature verified server-side via HMAC; wallet/order only updated after verified capture, not client claim	P: High C: High
PAY-02	As a Student, I want a digital wallet to top up and pay from, so that checkout is faster on future orders.	Top-up via Razorpay; balance debit uses MongoDB session transaction (atomic, no partial state)	P: Medium C: High
PAY-03	As a Student, I want to split an order's cost with friends, so that group print jobs are paid fairly.	Split request sent to friends; each accepts/declines; order proceeds only when all accept or creator covers remainder	P: Medium C: High
PAY-04	As a Student, I want to download a receipt for any completed order, so that I have proof of payment.	PDF receipt generated on demand; includes token number, cost breakdown, payment method	P: Low C: Low

6. Notifications

ID	User Story	Acceptance Criteria	Details
NOTIF-01	As a Student, I want push notifications for order status changes, so that I don't have to keep the app open.	Web Push via VAPID keys; falls back to in-app if permission denied	P: High C: Medium
NOTIF-02	As a Faculty registrant, I want the Admin to be notified instantly when I request role approval, so that my access isn't delayed unnecessarily.	Admin notification created + emitted via socket on registration	P: Medium C: Low
NOTIF-03	As any user, I want a notification centre showing read/unread status, so that I can review past alerts.	Paginated list, mark-as-read, filter by type (order/complaint/system)	P: Medium C: Low

7. Complaints

ID	User Story	Acceptance Criteria	Details
COMP-01	As a Student, I want to raise a complaint about print quality on a specific order, so that the issue gets resolved.	Complaint allowed only for orders in Processing/ReadyForPickup/Completed/Partial; category + description required	P: High C: Medium
COMP-02	As a Student, I want to chat with staff/admin inside a complaint thread, so that I can clarify details in real time.	Real-time messaging via socket rooms; message history persisted	P: Medium C: Medium
COMP-03	As Counter Staff/Admin, I want to update complaint status (Open → In Progress → Resolved/Closed), so that resolution is tracked.	Status change logged with actor + timestamp in statusHistory; student notified on each change	P: High C: Low

8. Guest Kiosk

ID	User Story	Acceptance Criteria	Details
GUEST-01	As a Guest (walk-in, non-registered), I want to authenticate with an email OTP, so that I can place an order without creating an account.	OTP rate-limited (5/email/30min, 10/IP/15min); session valid until browser closed or timeout	P: High C: Medium
GUEST-02	As a Guest, I want the same document upload and AI analysis experience as registered users, so that I get consistent quality checks.	Guest upload pipeline identical to registered user pipeline; flag as isGuest	P: High C: Medium
GUEST-03	As a Guest, I want to pay only at the counter (cash), so that I don't need a payment account.	Payment method locked to Cash; priority always fixed at level 2; staff sees "Guest" tag on queue	P: High C: Low
GUEST-04	As a Guest, I want to receive all order updates via email since I'm not logged into the app, so that I know when to collect my order.	All lifecycle notifications sent to guest auth email only, no push/socket for guests	P: Medium C: Low

9. Chat, Friends & Split Payment

ID	User Story	Acceptance Criteria	Details
CHAT-01	As a Student, I want to send friend requests to classmates, so that I can chat and split print orders with them.	Search by name/register number; request pending until accepted; duplicate requests blocked	P: Medium C: Medium
CHAT-02	As a Student, I want one-to-one and group chat with friends, so that I can coordinate group print jobs.	Real-time messaging (text/voice/media); read receipts; typing indicators	P: Low C: High
CHAT-03	As a Student in a group chat, I want to see a split payment card with live status per participant, so that I know who has paid.	Card shows paid/pending/declined per person; resend option for pending requests	P: Medium C: Medium

10. Document Tools (Free Utilities)

ID	User Story	Acceptance Criteria	Details
TOOLS-01	As a Student, I want to convert, resize, or compress a document/image without placing an order, so that I can prep files before printing elsewhere too.	No cost, no order created; tools work standalone from a dedicated page	P: Low C: Medium

11. Staff Dashboard

ID	User Story	Acceptance Criteria	Details
STAFF-01	As Counter Staff, I want a live, priority-sorted queue view, so that I process the correct order next.	Queue auto-updates via socket; sorted by priority score then creation time	P: High C: Medium
STAFF-02	As Counter Staff, I want to lock an order while I'm working on it, so that another staff member doesn't process it simultaneously.	Lock times out after 5 minutes of inactivity; locked orders show staff name to others	P: Medium C: Medium
STAFF-03	As Counter Staff, I want to verify a pickup OTP before marking an order Collected, so that only the rightful owner receives it.	OTP bcrypt-hashed and compared; wrong OTP attempts limited; success transitions to Completed	P: High C: Medium
STAFF-04	As Counter Staff, I want to collect cash payments and mark them confirmed, so that Cash orders enter the queue correctly.	Cash_Pending → Cash_Collected transition creates Payment record; order enters queue only after confirmation	P: High C: Low
STAFF-05	As Counter Staff, I want to report low stock or equipment faults, so that Admin can act before it disrupts service.	Report includes category, urgency, optional linked inventory item; Admin notified immediately if Urgent	P: Medium C: Low

12. Admin Dashboard

ID	User Story	Acceptance Criteria	Details
ADMIN-01	As an Administrator, I want to configure service pricing and time-per-page values, so that costs reflect real operating expenses.	Pricing changes logged in immutable ActivityLog; changes apply only to new orders, never retroactively	P: High C: Medium
ADMIN-02	As an Administrator, I want to approve or reject Faculty role requests, so that only legitimate staff get elevated access.	Approve sets role=Faculty; reject reverts pendingRole to null with notification to user	P: High C: Low
ADMIN-03	As an Administrator, I want a dashboard of daily orders, revenue, and open complaints, so that I can monitor operations at a glance.	Stats refresh on load; queue counts per service type shown live	P: High C: Medium
ADMIN-04	As an Administrator, I want to view an immutable audit log of all sensitive actions, so that I have accountability for pricing/user changes.	ActivityLog blocks all update/delete operations at the schema level; filterable by action type and date	P: High C: Medium
ADMIN-05	As an Administrator, I want to manage inventory thresholds and view forecasted depletion, so that supplies never run out unexpectedly.	Below-threshold items trigger notification to Staff + Admin; usage rate calculated from consumption history	P: Medium C: Medium
ADMIN-06	As an Administrator, I want to force-open or force-close the shop outside scheduled hours, so that I can handle holidays or emergencies.	Manual override persists until reverted to "follow schedule"; cron respects manual mode and doesn't overwrite it	P: Medium C: Low
ADMIN-07	As an Administrator, I want to deactivate a user account, so that I can block misuse without deleting their history.	isActive=false blocks login; existing order/payment records remain intact	P: Medium C: Low

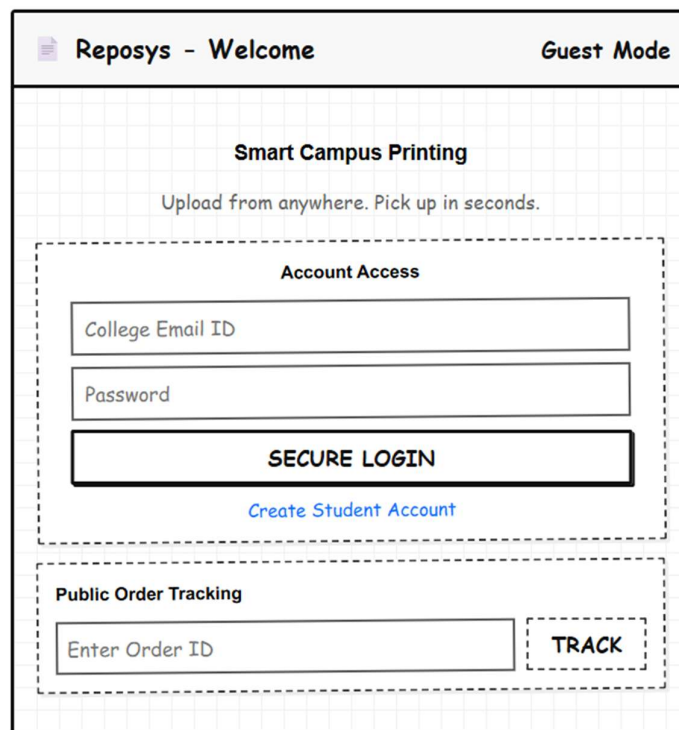
ADMIN-08	As an Administrator, I want to export analytics reports (CSV/PDF) by date range, so that I can present data for academic evaluation or planning.	Export covers daily orders, revenue, service breakdown, payment breakdown; respects selected date filter	P: Low C: Medium
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3. Proposed UI/UX System Design

The User Interface is planned to be constructed as a Single Page Application (SPA) using React.js. The UI/UX methodology prioritizes a "Mobile-First" progressive design to ensure usability across mobile devices via the PWA architecture. Below are the structural CSS wireframes proposing the layouts for each of the four primary roles.

3.1 Guest Interface: Landing & Authentication

The Guest interface will act as the digital storefront, providing clear calls to action to register or log in, alongside a public tracking feature for unauthenticated status checks.



Reposys - Welcome Guest Mode

Smart Campus Printing
Upload from anywhere. Pick up in seconds.

Account Access

College Email ID

Password

SECURE LOGIN

[Create Student Account](#)

Public Order Tracking

Enter Order ID **TRACK**

Figure 3.1: Proposed Guest Landing & Authentication Wireframe

3.2 User Interface: The Document Lab

The User dashboard will be designed as an interactive "Wizard". Users will be visually guided through a step-by-step pipeline: File Upload → Parameter Selection → Cost Calculation.


Document Lab

Wallet: ₹145.00

Step 1: Upload PDF


Drag & Drop Document Here
or tap to browse (Max 50MB)

Step 2: Print Configuration

Black & White

Color

Copies:

1

Calculated Cost:

₹15.00

Based on 10 pages B&W.

PAY & SUBMIT TO QUEUE →

Figure 3.2: Proposed User Interface (Mobile-First Document Upload)

3.3 Staff Interface: Live Fulfillment Dashboard

The Staff module will prioritize high-contrast visibility for queue monitoring and rapid-click action buttons, allowing counter staff to process physical papers without wrestling with a complex digital UI.

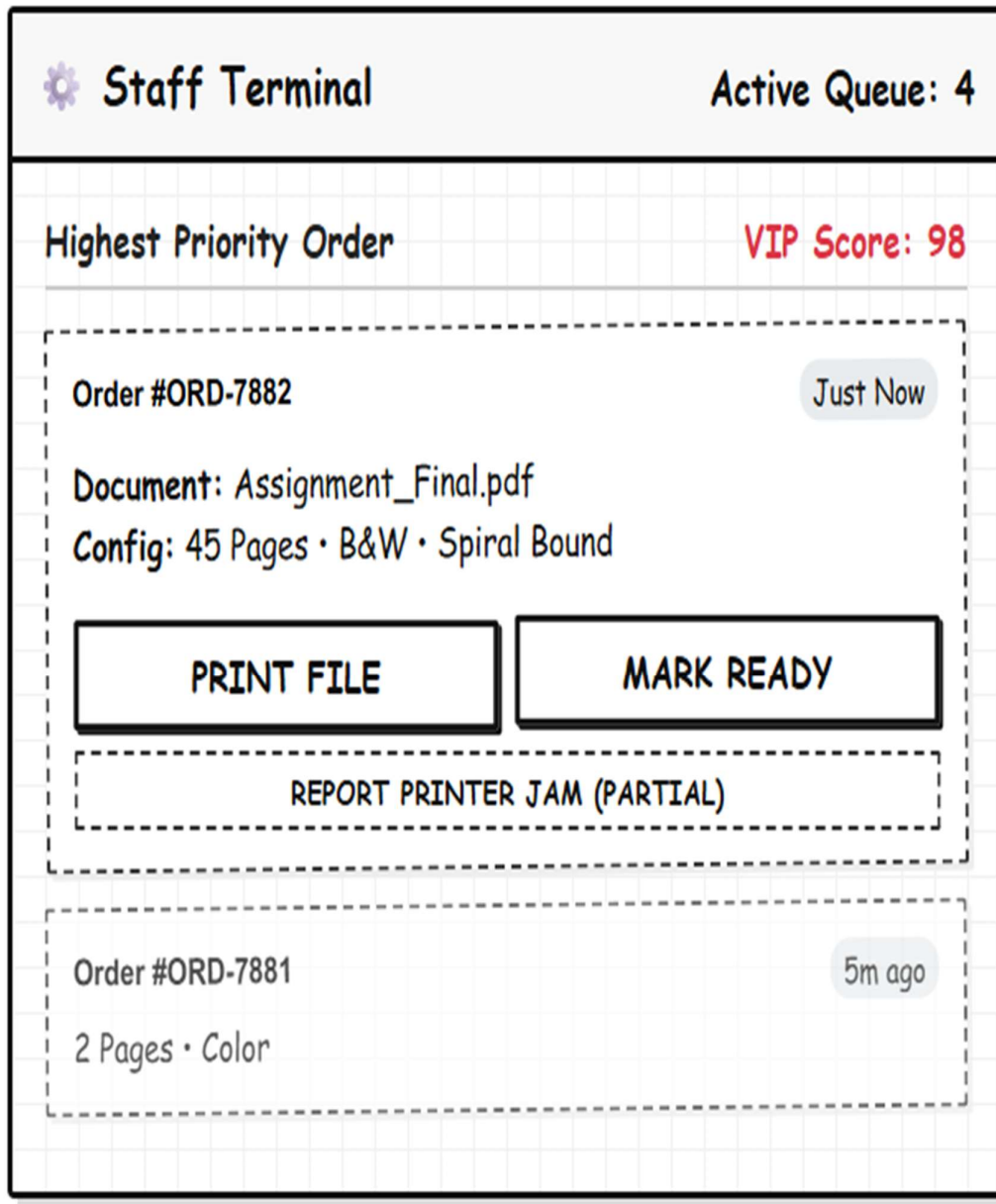


Figure 3.3: Proposed Staff Interface (Order Queue & Fulfillment)

3.4 Admin Interface: Global Oversight

The Admin interface will be data-dense, designed for desktop monitors but responsive for tablets. It will feature Demand Forecasting cards and system-wide configuration toggles.


Admin Command Center

System Health: Optimal

TODAY'S REVENUE
₹4,250

PAGES PRINTED
1,420


Inventory Forecast Alert

Based on the current 7-day rolling average, **A4 Paper (Bundle)** is projected to run out in approximately **36 hours**.

Pricing Configuration

A4 B&W Base Price

₹1.50

A4 Color Base Price

₹5.00

UPDATE SYSTEM PRICING

Figure 3.4: Proposed Admin Interface (Analytics & System Config)

3.5 User Interface: Wallet & Split Payments

A FinTech-inspired digital wallet interface that allows students to top up via Razorpay and send peer-to-peer split payment requests for group assignments.

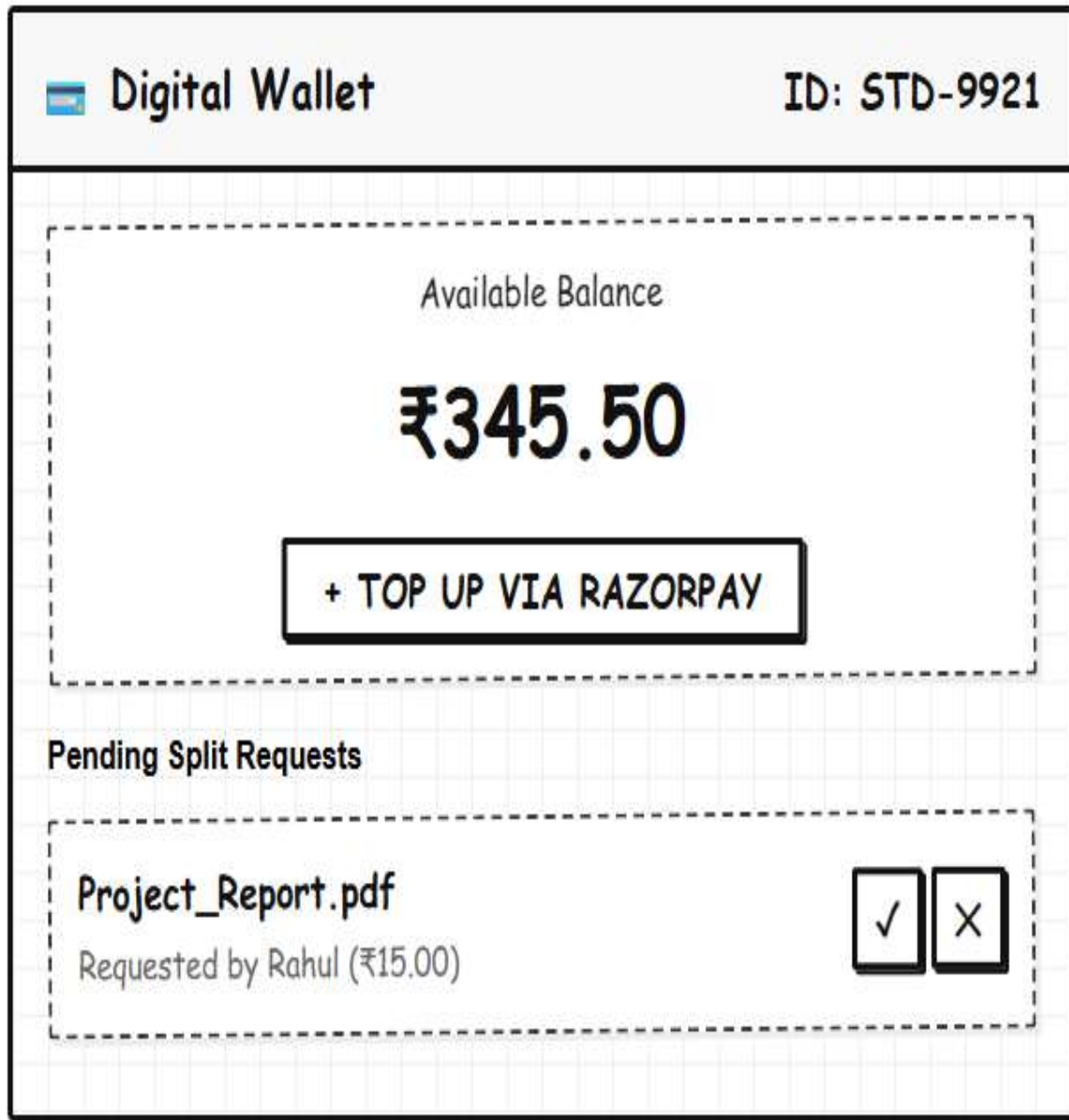


Figure 3.5: Proposed Wallet & Peer-to-Peer Split Payment Wireframe

3.6 Universal Interface: Live Order Tracker

A WebSocket-powered timeline view that updates in real-time as the counter staff processes the order. Accessible by both authenticated users and guests (via public link).

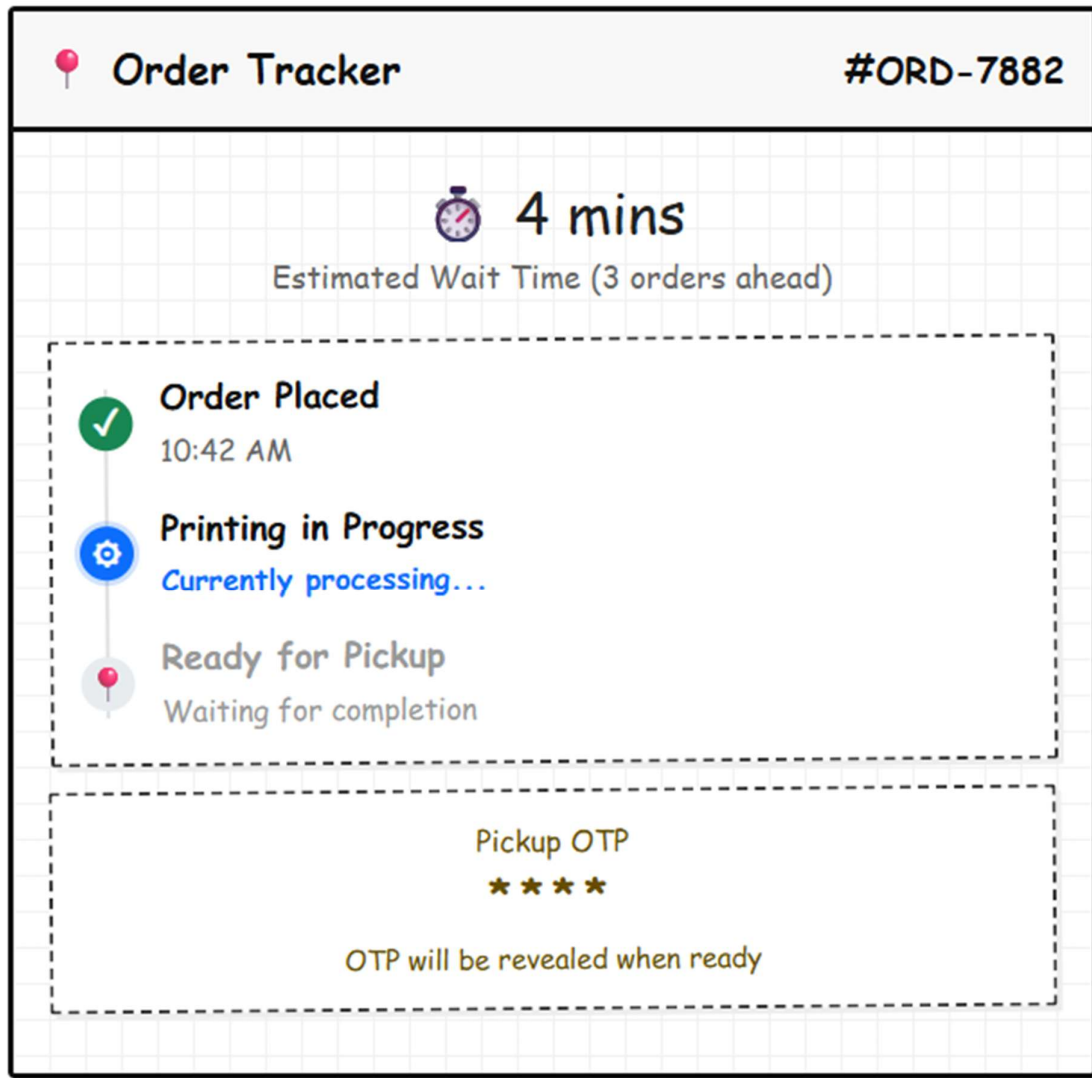


Figure 3.6: Proposed Live Order Tracking Timeline Wireframe

3.7 AI Assistant Chat Widget

A floating context-aware Gemini AI chatbot capable of answering queries based on live system variables (e.g., current shop hours, queue length, and pricing).

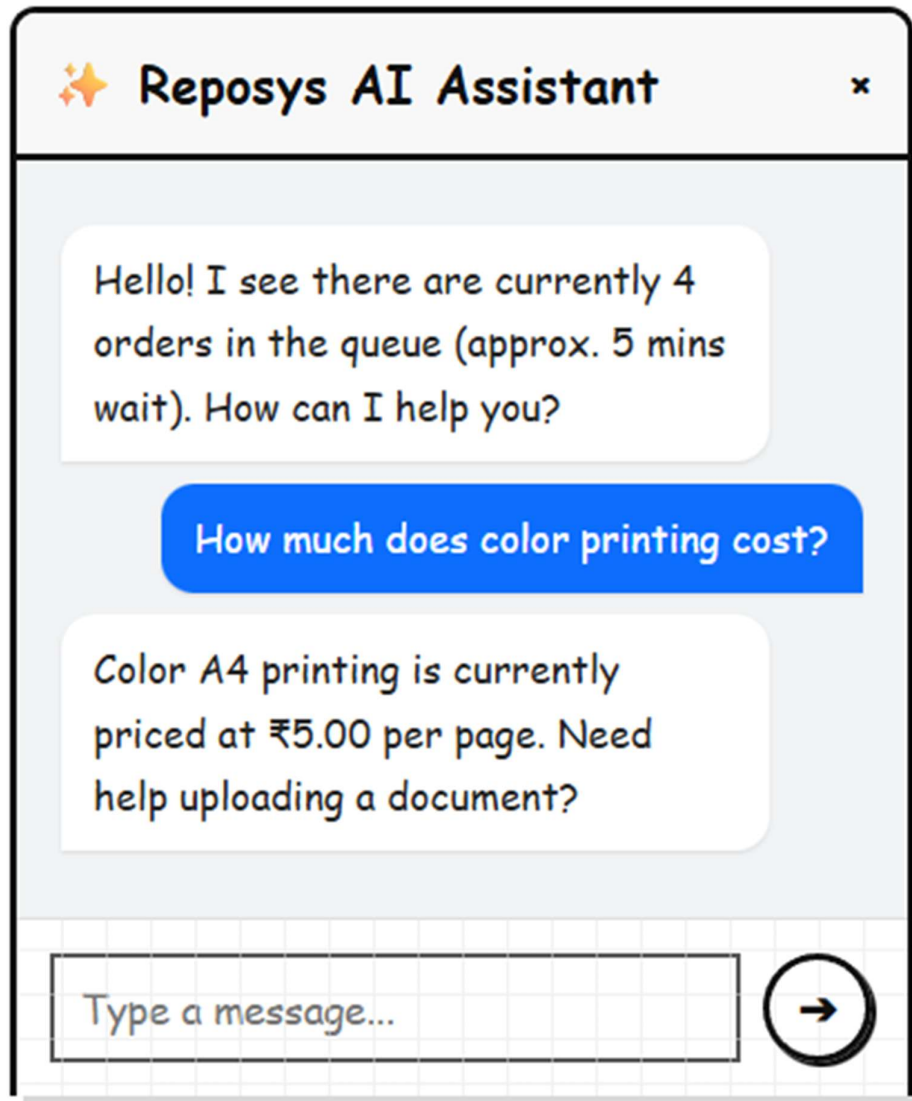


Figure 3.7: Proposed Context-Aware AI Chatbot Interface